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Dear ELLIS Committee,

Re: Application for the PhD Program at ELLIS Institute Finland

I am writing to apply for the PhD Program at ELLIS Institute Finland. My research has increasingly focused on understanding how intelligent systems, particularly Large Language Models, can operate effectively in tasks that require social awareness and an understanding of human behaviour. While recent benchmarks such as ARC-AGI have advanced the study of learning, reasoning, and generalization in AI, I have been interested in exploring how intelligence manifests in social settings: the ability to interpret other people, infer their intentions, adapt to social context, and navigate interactions where participants may have different goals and incomplete information. These abilities constitute core aspects of Theory of Mind: the capacity to reason about the beliefs, intentions, and perspectives of others. Studying these abilities in AI systems may not only help us build more capable and socially aware systems, but also provide new ways of thinking about the nature of intelligence and the capabilities that contribute to human intelligence. My research therefore focuses on computationally modelling these abilities in LLMs, evaluating how reliably they emerge, and understanding how to make such systems more trustworthy, drawing on ideas from machine learning, cognitive science, and the social sciences.

My curiosity about socially beneficial AI began with my work under Professor Emily Aiken, where I developed a unified social-harm and risk-detection framework: a multiclass machine-learning system for identifying ten categories of online risk, including mental-health distress and cyberbullying. The framework was trained on public social-media posts from Reddit and Twitter, drawing on datasets with human-annotated and weakly or semi-supervised labels. Rather than treating model outputs as clinical judgments, we used them to identify linguistic patterns associated with different forms of online harm. However to examine whether these predictions were grounded in meaningful signals, I combined SHAP attribution with an LLM that translated influential features into human-readable explanations. This work, which received a Best Paper Award at AAAI-26's AI in Medicine and Healthcare program, was my first experience of seeing how machine learning could be used to model socially meaningful behaviour. What stayed with me, however, was not just the technical challenge of classification, but the ambiguity behind the task: the same sentence can be harmless banter in one setting and a serious risk signal in another. The datasets made this challenge especially clear: labels were attached to isolated posts, even though mental-health experiences and cyberbullying can co-occur and often require broader social or longitudinal context for interpretation. This made it clear to me that social context is not an optional layer added after modelling; it is often the condition for interpreting language correctly and deploying AI responsibly in high-impact domains.

Following this work, I became interested in computational humor as a particularly clear setting for studying social and cognitive abilities in LLMs. Humor requires more than producing linguistically plausible language through the LLM objective of predicting the next most probable token: it can depend on incongruity, surprise, contextual knowledge, shared assumptions, and an understanding of what another person may find amusing. I therefore developed the Cognitive Synergy Framework (CSF), a cognitively motivated approach that translates psychological theories of humor into specific generative personas such as a *Cynic* focused on hypocrisy and social critique, an *Observer* mapping unwritten social norms, and a *Neurotic* leveraging internal anxiety and overthinking to generate significantly more diverse training data. I then used this data to distill humor generation abilities into smaller LLMs through supervised fine-tuning and reinforcement learning, resulting in the HumorGen model collection. HumorGen-7B variants consistently outperformed substantially larger open-weight baselines on held-out and out-of-distribution benchmarks. More importantly, this work strengthened my interest in using cognitive and behavioural insights not only to improve AI capabilities, but also to understand what these capabilities require. The work was accepted to the EMNLP 2026 Main Conference, and I subsequently extended the framework to multilingual humor generation through the CLEF 2026 JOKER task, where my submissions achieved first positions in English, French, and Spanish. The resulting work was accepted to CLEF 2026 Conference.

As my work progressed, I became increasingly interested not only in models with these abilities, but also in how reliably we can evaluate them. Following HumorGen, I developed HumorRank to study the

reliability of LLM-based evaluation for humor generation. With extensive research, we found that standard LLM-as-a-judge approaches fail in different ways when it comes to evaluating humor such as: score anchoring, model family bias that make it difficult to distinguish between candidate outputs produced by different humor generation models. We therefore developed HumorRank, a tournament-based evaluation framework using Bradley–Terry estimation to produce more stable comparisons across judges. Beyond ranking models, the framework also allowed me to examine the kinds of errors they make, revealing structured failure patterns such as Cliché and Weak Punchlines, while specialized humor generation models showed a distinct pattern of overexplained jokes. This work shifted my interest further toward evaluation: as LLMs acquire increasingly complex social and cognitive abilities, the challenge is not only to measure whether they perform well, but to understand whether the underlying capability is reliable and where it breaks down. I am now extending this direction to the evaluation of social interaction quality in LLMs.

Moving into the domain of evaluating social intelligence in LLMs, my recent work explores how agents navigate high-friction environments where objectives may not be fully aligned. I led the development of SIDE-Bench, a human-validated social interaction benchmark comprising 150 scenarios across domains such as commercial negotiations, interpersonal dilemmas, and customer service. Unlike standard benchmarks, success in SIDE-Bench requires models to balance private, verifiable sub-goals with the need to maintain interpersonal rapport while interacting with a resistant partner. To understand why agents fail in these settings, I developed the Conversation-Partner Intent Tracking (C-PIT) framework, which evaluates Theory of Mind by tracking an agent's beliefs about its interaction partner as they evolve turn by turn. Our evaluation of seven frontier and open-weight models, currently under review at ICLR 2027, found that models can correctly track their partner's objectives while still producing actions that violate those inferred constraints. This action-belief disconnect suggests that strong reasoning about another agent does not necessarily translate into socially appropriate action. This work has brought together the questions that have emerged throughout my research: how can we build systems that understand other agents, adapt to social context, and act reliably when goals, information, and constraints are not fully aligned?

Looking ahead, I want to explore three interconnected areas: social intelligence and human-AI interaction, multi-agent systems and LLM reasoning, and the evaluation and alignment of increasingly capable AI systems. Within social intelligence and human-AI interaction, I am interested in how AI agents reason about people, interpret their intentions and perspectives, and how interactions with AI systems shape human agency and sense of control. Within multi-agent systems and LLM reasoning, I am interested in how agents reason about other agents, adapt to changing interaction contexts, and operate when objectives, preferences, and information are not fully aligned. Finally, in evaluation and alignment, I am particularly interested in developing reliable methods for evaluating these capabilities and understanding when increasingly capable systems behave in ways that remain consistent with human objectives and preferences. Across these areas, I hope to investigate how advances in LLM reasoning and multi-agent learning can support AI systems that interact effectively with humans and other agents while remaining robust, trustworthy, and responsive to the people they affect.

This is what particularly attracts me to the PhD Program at ELLIS Institute Finland: the opportunity to pursue these questions within a research environment spanning machine learning, natural language processing, human-AI interaction, and AI for health. I am particularly drawn to the breadth of expertise across the ELLIS network, especially Professors Samuel Kaski, Qi Chen, Kristiina Jokinen, and Eric Malmi. The possibility of working with Prof. Kaski on collaborative AI, human-AI interaction, and Theory of Mind is particularly exciting to me. With Prof. Chen, I would be interested in exploring trustworthy and continually adapting AI systems, particularly how AI agents can generalize and adapt as their environments, information, and objectives change. I would also be interested in exploring dialogue and human-agent interaction with Prof. Jokinen. With Prof. Malmi, I see opportunities to explore LLM reasoning, agentic capabilities, and interdisciplinary applications of language models. The program's emphasis on interdisciplinary collaboration and connections across academia and industry would also allow me to develop these ideas beyond individual benchmarks and toward AI systems that can be evaluated in realistic human settings. I see ELLIS as an environment where I can bring my experience in LLMs, social interaction, and evaluation into conversation with complementary perspectives across machine learning and human-centred AI, while developing my research direction with the guidance of faculty whose expertise aligns with the specific questions I pursue.

Sincerely,
Edward Ajayi